

Galaxy-Pick

PJ1 · Samsung Galaxy Phone Recommendation System — Team Execution Plan

Module: Samsung GenAI Capstone (PJ1) · Team: Member 1–4 · Build window: 3–5 days · Date: 14 Jul 2026

The decision, in one line. Build a **deterministic** Galaxy phone recommender — a curated 2024–2026 dataset → four transparent **0–10 scores** → a **Weighted Sum Model** → an explained **top-3** — delivered as a **Jupyter notebook** (the graded core) plus a **Samsung-styled Streamlit app**, with an **optional, free-tier Gemini** layer for natural-language input that **always falls back** to an offline rule-based parser. It runs with no API key and no network.

Executive summary (one page — read this first)

What we're building

- Notebook (graded):** EDA, the four scores with documented logic, 4 personas, the WSM + a worked example.
- Streamlit app (demo):** pick a persona or type a need → ranked top-3 cards with a score breakdown and a one-line "why."
- One AI feature:** natural-language persona parsing (Gemini optional; deterministic fallback always on).
- Responsible-AI panel:** shows real vs. mock data and calls scores "heuristic."

Why this wins

PJ1 is graded on a small, specific rubric (personas, 0–10 scores, WSM, EDA findings, a UI, a presentation) and on applying **Ch.2 prompt engineering**, **Ch.3 model selection**, and **Ch.4 Responsible AI**. The way to stand out is to **nail the rubric flawlessly** and add **one** on-theme differentiator — not to build production infrastructure.

Deliberately cut (they score nothing here and signal "misread the assignment"):

Fine-tuning

RAG

Embeddings

Vector DB

Backend / DB / Auth

Grok (paid)

Hugging Face

Runtime scraping

~36
curated models (2024–2026)

4
scores & personas

₹0
cost — 100% free/OSS

0
API keys required to run

Roadmap (5 phases)

Phase	Focus	Effort
0	Setup + skeleton + env	0.5d
1	Curated dataset + scores	1d
2	Notebook — graded core	1d

Team split

Seat	Owns
Member 1 — Data	Dataset, cleaning, provenance
Member 2 — Engine	Scoring, WSM, notebook, tests

3	Streamlit app	1d
4	NL parsing + Responsible AI	0.5–1d
5	Tests + polish + slides	0.5–1d

Member 3 — UI	Streamlit app, Samsung theme
Member 4 — AI/Present	NL parse, explanations, slides

Stack: Python 3.11 · pandas · Jupyter · Streamlit · matplotlib · pytest · (optional) Gemini.

1 · Project understanding

- **Objective:** a shopping-assistant prototype that recommends the right Galaxy phone for a shopper and **explains why**, ranked.
- **Users:** Samsung Mobile e-commerce shoppers, modelled as 3–4 personas; evaluated by the instructor / Samsung R&D engineers.
- **Graded deliverables:** personas with weights summing to 1.0 · a cleaned mock/curated dataset · 3 EDA findings · four 0–10 scores with documented logic · the WSM + one worked example · a working UI screenshot · a short presentation.
- **Evaluation:** Mini Project 80% + Presentation 20% — compliance with the named steps and visible use of Ch.1–4, *not* infrastructure.
- **Hidden traps:** scoring can look arbitrary (make it transparent); GenAI data has realism traps; Ch.4 wants you to *show* where AI was guarded; over-scoping burns time and signals misunderstanding.

2 · Gap analysis

Have	Missing	Biggest risk
PJ1 chosen; empty repo; real S/Z slice of data	Clean canonical dataset (1 row/model)	Over-engineering vs. the rubric
Two Kaggle scrapes (2022–23 — reference only)	The four 0–10 scores + WSM	Live-API failure at demo → neutralised by fallback
The brief's WSM spec	Personas, Streamlit UI, notebook, slides	Polishing UI before the core works

De-risking rule: finish the graded **notebook** by end of Phase 2 — after that you're pass-safe and everything else is upside.

3 · Dataset strategy

Verdict on the source data: the S-/Z-series rows in the team CSV are real and good — keep them. The A/M/F rows (A12, A14...) are **fabricated arithmetic padding** with wrong labels — **drop them**. The Kaggle scrapes are 2022–23, so with a 2024–2026 catalog they no longer match — demote to a **calibration reference** only.

Key insight — more phones does **not help.** A WSM has no training step, so extra rows can't raise accuracy — they only add noise. **Curate, don't accumulate.**

Recommended architecture — one row per model:

- **Range:** 2024–2026. 2024–25 real; every **2026 row is `spec_source=mock`** (a projection), flagged honestly.
- **Size:** ~36 models (27 real / 9 mock) — spans budget → flagship → foldable.
- **Images:** free, offline **placeholder cards** drawn in code (zero copyright risk).
- **Columns:** identity · raw specs · four scores · `spec_source` .

The curated dataset (36 models)

Model	Series	Yr	₹ Price	RAM	Chipset	Cam	mAh	Seg	Src
Galaxy S24	S	2024	87,559	8	Exynos 2400	50	4000	business	real

Galaxy S24+	S	2024	1,07,559	8	Exynos 2400	50	4900	business	real
Galaxy S24 Ultra	S-Ultra	2024	1,47,559	12	SD 8 Gen3	200	5000	photography	real
Galaxy S24 FE	S-FE	2024	67,559	8	Exynos 2400e	50	4700	gaming	real
Galaxy Z Flip6	Z-Flip	2024	1,17,559	8	SD 8 Gen3	50	4000	business	real
Galaxy Z Fold6	Z-Fold	2024	1,72,559	8	SD 8 Gen3	50	4400	business	real
Galaxy S25	S	2025	88,559	8	SD 8 Elite	50	4000	business	real
Galaxy S25+	S	2025	1,07,559	8	SD 8 Elite	50	4900	business	real
Galaxy S25 Ultra	S-Ultra	2025	1,47,559	12	SD 8 Elite	200	5000	photography	real
Galaxy Z Flip7	Z-Flip	2025	1,22,559	8	SD 8 Elite	50	4300	business	real
Galaxy Z Fold7	Z-Fold	2025	1,82,559	8	SD 8 Elite	200	4600	business	real
Galaxy A15 5G	A	2024	14,999	6	Dimensity 6100+	50	5000	budget	real
Galaxy A25 5G	A	2024	19,999	8	Exynos 1280	50	5000	budget	real
Galaxy A35 5G	A	2024	26,999	8	Exynos 1380	50	5000	business	real
Galaxy A55 5G	A	2024	35,999	8	Exynos 1480	50	5000	business	real
Galaxy M15 5G	M	2024	12,999	6	Dimensity 6100+	50	6000	budget	real
Galaxy M35 5G	M	2024	18,999	8	Exynos 1380	50	6000	budget	real
Galaxy M55 5G	M	2024	26,999	8	SD 7 Gen1	50	5000	gaming	real
Galaxy F15 5G	F	2024	12,499	6	Dimensity 6100+	50	6000	budget	real
Galaxy F55 5G	F	2024	26,999	8	SD 7 Gen1	50	5000	gaming	real
Galaxy A16 5G	A	2025	16,999	6	Dimensity 6300	50	5000	budget	real
Galaxy A26 5G	A	2025	21,999	8	Exynos 1380	50	5000	budget	real
Galaxy A36 5G	A	2025	26,999	8	SD 6 Gen3	50	5000	business	real
Galaxy A56 5G	A	2025	35,999	8	Exynos 1580	50	5000	business	real
Galaxy M16 5G	M	2025	13,999	6	Dimensity 6300	50	6000	budget	real
Galaxy M56 5G	M	2025	27,999	8	Exynos 1580	50	5000	gaming	real
Galaxy F16 5G	F	2025	13,499	6	Dimensity 6300	50	6000	budget	real
Galaxy S26	S	2026	89,999	12	SD 8 Elite Gen2	50	4300	business	mock
Galaxy S26+	S	2026	1,09,999	12	SD 8 Elite Gen2	50	4900	business	mock
Galaxy S26 Ultra	S-Ultra	2026	1,39,999	12	SD 8 Elite Gen2	200	5200	photography	mock
Galaxy S26 FE	S-FE	2026	62,999	8	Exynos 2500	50	4900	gaming	mock
Galaxy Z Flip8	Z-Flip	2026	1,19,999	12	SD 8 Elite Gen2	50	4400	business	mock
Galaxy Z Fold8	Z-Fold	2026	1,84,999	12	SD 8 Elite Gen2	200	4800	business	mock
Galaxy A57 5G	A	2026	37,999	8	Exynos 1680	50	5000	business	mock
Galaxy A17 5G	A	2026	17,999	6	Dimensity 6400	50	5000	budget	mock
Galaxy M57 5G	M	2026	28,999	8	Exynos 1580	50	5000	gaming	mock

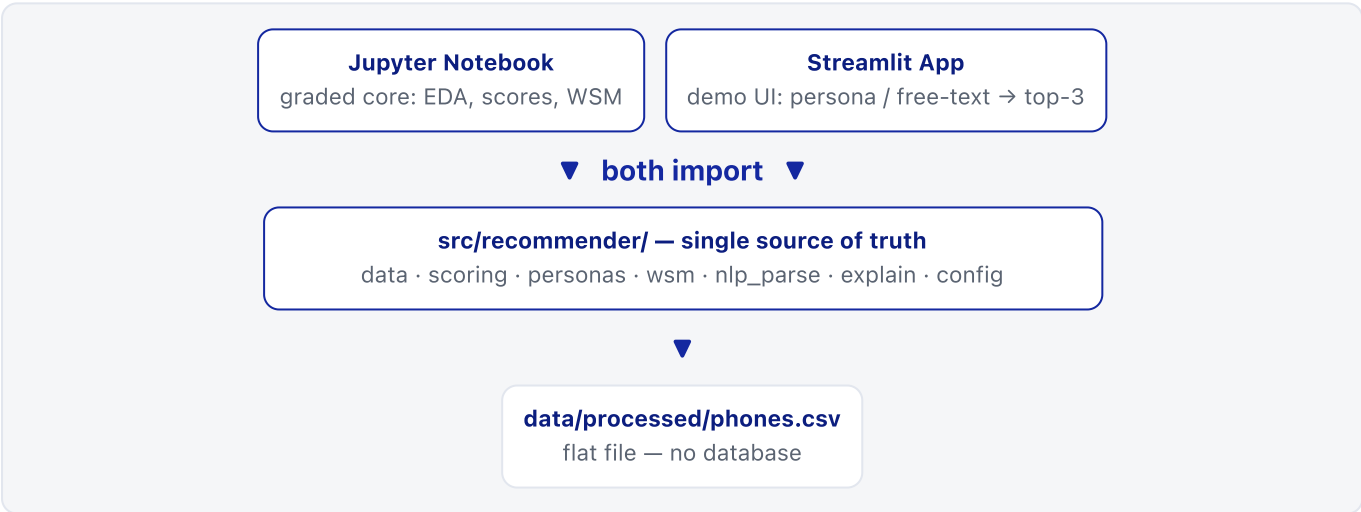
Full CSV + exact scoring formulas live in [PLAN.md](#) (Appendix A & B). Prices are representative base MRP in INR.

4 · AI / ML strategy

Question	Decision	Why
Core logic	Weighted Sum Model	Required by brief; fully explainable — the property that wins.
Fine-tuning	No	No training data or step; signals misread assignment.
RAG / Vector DB	No	36 rows fit in a DataFrame; nothing to retrieve.
Embeddings	No	WSM is more explainable than a similarity black box.
LLM API	Gemini — 1 job, optional	Parse free-text → weights (Ch.2/3/4). Free tier; always has a fallback.
Grok / HF	No	Paid / off-theme / extra keys, zero added value.

The one standout AI feature. A free-text box ("a phone for photography under ₹50k") → Gemini returns JSON weights *if a key exists*, else a deterministic keyword parser does the same. Output is validated before use. Showcases Ch.2–4 **and** guarantees the demo can't break.

5 · System architecture (right-sized)



No backend, no DB, no auth, no cache. Streamlit calls in-process Python (microseconds). Gemini is the only network call — wrapped in try/except with a fallback. Deploy = local `streamlit run` ; optional free Streamlit Community Cloud.

6 · Feature versions (A / B / C)

Version	Scope	Wow
A · Minimal (~1d)	15 phones, scores, WSM, 4 personas, ipywidgets dropdown. Passes 100% of the rubric.	—
B · Competitive ★ (3–5d)	~36 curated models, Streamlit UI, NL persona parsing + fallback, explainable breakdown, Responsible-AI panel.	Type in plain English → explained ranked result; works offline.
C · Award-ready (B + extras)	Live weight sliders, contrastive "why not X", close-call flag, side-by-side compare.	Tune priorities live & watch the ranking move.

Plan of record: build **B**; pull the weight sliders + close-call flag from C in only if Phase 5 is green.

7 · UI / UX strategy

- **Layout:** sidebar (persona dropdown + free-text + budget slider) → main: top-3 cards → ranked table → score-breakdown bar.
- **Samsung language:** Samsung Blue #1428A0 , One-UI rounded cards, generous whitespace, clean sans type.
- **Accessibility:** strong contrast, alt text, never color-only meaning (always show the number), keyboard-navigable.
- **On Stitch:** fine as a mockup aid, but don't adopt a full React stack — it abandons the required Jupyter environment. Streamlit gives better polish-per-hour.

8 · Development roadmap

Phase	Objective	Key files	Done when
0 · Setup	Env + skeleton	environment.yml, folder tree	conda env builds; streamlit runs
1 · Data	Canonical dataset	build_dataset.py, phones.csv	~36 rows, all asserts pass
2 · Notebook	Graded core	pj1_analysis.ipynb	Runs top-to-bottom; example = 7.5 → pass-safe
3 · App	Streamlit UI	src/recommender/*, streamlit_app.py	Persona → explained top-3 in browser
4 · AI + RAI	NL parse + guardrails	nlp_parse.py, explain.py	Works with and without a key
5 · Polish	Tests, theme, slides	tests/, presentation/	pytest green; deck rehearsed

Team split — day by day

Day	Member 1 · Data	Member 2 · Engine	Member 3 · UI	Member 4 · AI/Present
1 (P0–1)	Drive dataset	All hands verify specs · freeze function signatures		
2 (P2)	EDA + provenance	Notebook core → pass-safe	App skeleton + theme	Persona copy + fallback parser + slide outline
3 (P3–4)	Edge cases + screenshots	Wire src/ + tests	Finish app	NL parse + explain + RAI panel
4 (P5)	Integrate + tests; close-call flag (M2)		Weight sliders (stretch)	Deck + demo script
5	Rehearse · fallback drill · final screenshots · finalize deck			

9 · Project folder structure

```
galaxy-pick/
├─ CLAUDE.md · PLAN.md · README.md
├─ environment.yml · requirements.txt · .env.example
├─ .streamlit/config.toml           # Samsung theme
├─ data/{raw, processed}/phones.csv
├─ scripts/build_dataset.py         # raw → scored CSV
├─ notebooks/pj1_analysis.ipynb    # GRADED CORE
├─ src/recommender/                # single source of truth
│   └─ config · data · scoring · personas
│   └─ wsm · nlp_parse · explain · cards
├─ app/streamlit_app.py             # demo UI
├─ assets/images/                  # empty → code draws placeholder cards
├─ tests/{test_scoring, test_wsm}.py
└─ presentation/                   # slides + screenshots
```

Deliberately no backend/ , db/ , or auth/ .

10 · Technology stack

Python 3.11 (Anaconda)

pandas

numpy

Jupyter

Streamlit

matplotlib

pytest

python-dotenv

google-generativeai (optional)

Cost check: every component is free/open-source and runs locally. Gemini, if used, is Google AI Studio's free tier; the deterministic default needs no key and no network. No paid services anywhere.

11 · Risk analysis & mitigation

Risk	Mitigation
Gemini key absent / fails at demo	Deterministic fallback <i>is</i> the core; live layer is optional, wrapped in try/except.
Judge spots fabricated names	Drop the arithmetic A/M/F rows; <code>spec_source</code> provenance; honest "mock where noted."
2026 specs aren't real	Every 2026 row flagged mock; presented as a projection in the panel + slides.
Scoring seen as arbitrary	Document every formula; show the weight sliders / sensitivity; call scores "heuristic."
Scope creep (RAG/DB/backend)	Freeze at Version B; C items only if Phase 5 is green.
Polishing before core works	Notebook done by end of Phase 2 → pass-safe before any UI polish.

12 · Creativity opportunities (on-theme, not gimmicks)

- **Live weight sliders** — reorder results in real time; makes the WSM self-evident in the room.
- **Contrastive "why not X"** — explains the phone it *didn't* pick (over budget / weaker on your top factor).
- **Close-call flag** — when the top two are within a small margin, say so (honest UX = Ch.4).
- **Provenance & limitations panel** — real vs. mock, scoring caveats; judges reward visible Responsible AI.

13 · Final recommendation

Build Version B: a deterministic Galaxy-Pick engine (curated 2024–2026 dataset → four documented 0–10 scores → WSM → explained top-3), as a Jupyter notebook (graded core) + a Samsung-styled Streamlit app, with an optional free-tier Gemini layer that always degrades to a deterministic parser.

Why over the alternatives: the big production stack scores nothing here, costs days you don't have, and signals you misread a prompt-engineering / Responsible-AI exercise. Version A merely passes; B adds the differentiators (natural-language input, explainability, a Responsible-AI story) within the 3–5 day window; C's flourishes drop into B *only if* time remains. Deterministic-core-first means the project is fully demonstrable with zero API dependency — Gemini becomes a clean bonus, never a single point of failure.

Verification & sign-off checklist

- ☐ `build_dataset.py` → ~36 rows (27 real / 9 mock); all asserts pass.
- ☐ Notebook *Restart & Run All* → no errors; worked example prints **7.5**; 3 EDA findings written.
- ☐ App: each persona → correct top-3; "photography under ₹50k" → camera-weighted, all $\leq$ ₹50k.
- ☐ **Fallback drill**: no key + Wi-Fi off → NL parse + explanations still work.
- ☐ `pytest -q` green; screenshots + deck ready.

Companion files in the repo: `CLAUDE.md` (coding-agent rules) and `PLAN.md` (full phase-by-phase build spec with all module code, the complete dataset CSV, scoring formulas, tests, and the Gemini-prompt appendix).